Contents

Chapter 1

Standard Model of Cosmology

Contents

Introduction

Cosmology is the study of the Universe, aiming to understand its origin and evolution. Within the broader landscape of science and physics, it occupies a unique position: it is one of the few fields where phenomena cannot be reproduced to directly test theoretical predictions. We have access to a single realization of the Universe, cannot replicate its underlying mechanisms, and can only observe a limited portion of its information.

This thesis aims to contribute to our understanding of the earliest moments of the Universe. This first chapter provides a structured overview of the standard cosmological model, known as ΛCDM, guiding the reader from its fundamental principles to its current limitations. The chapter is organized into five sections. The first introduces the key components of the model, including its foundations, assumptions, and limitations, while the second presents a qualitative overview of the thermal history of the Universe. The third focuses on the Cosmic Microwave Background (CMB), one of the most powerful probes of the early Universe. The fourth section presents the main sources of contamination affecting CMB observations. Finally, the fifth section reviews past CMB experiments and their major results, before discussing upcoming missions.

1.1 The Standard Model of Cosmology

Modern cosmology is built upon a set of simple but powerful principles that allow us to describe the large-scale behaviour of the Universe. Observations reveal that the Universe is expanding and appears remarkably homogeneous and isotropic on large scales. These properties provide the foundation for the standard cosmological model, known as ΛCDM.

In this section, we introduce the key elements of this framework. We begin with the observational evidence for the expansion of the Universe, then present the assumptions underlying its large-scale description. We subsequently describe the main components of the Universe, before discussing the limitations of this model and the role of inflation as a possible resolution.

1.1.1 Expanding universe

A major milestone in modern cosmology was the discovery that the Universe is expanding, independently established by Lemaître in 1927 [1] and Hubble in 1929 [2]. Their observations showed that distant galaxies recede from each other, with a velocity proportional to their distance. This relation, now known as Hubble-Lemaître law, provided the first evidence that the Universe is not static but evolving.

This expansion is described by the *scale factor* $a(t)$, which quantifies how distances between comoving points evolve with time. By convention, the scale factor is normalized to $a(t_0) = 1$ today. As the Universe expands, the scale factor increases, implying that physical distances between distant, gravitationally unbound objects grow proportionally with time, as illustrated in Figure 1.1.

An observable consequence of this expansion is the *cosmological redshift* z. As light propagates through an expanding Universe, its wavelength is stretched along with space. This effect is not a simple Doppler redshift [3], but rather a consequence of the expansion of spacetime, hence the name of cosmological redshift, illustrated in Figure 1.2.

The redshift is defined as:

$$1 + z \equiv \frac{\lambda_{\mathrm{obs}}}{\lambda_{\mathrm{emit}}} \equiv \frac{a_{\mathrm{obs}}}{a_{\mathrm{emit}}} = \frac{1}{a_{\mathrm{emit}}} \tag{1.1}$$

This relation allows us to infer distances and the expansion history of the Universe from observations of spectral lines. In his original work, Hubble measured the redshift of galaxies and compared it to their distances, establishing the linear relation between velocity and distance shown in Figure 1.3. This observation demonstrates that the expansion of the Universe is uniform on large scales: every observer sees distant galaxies receding, without implying a special position in space.

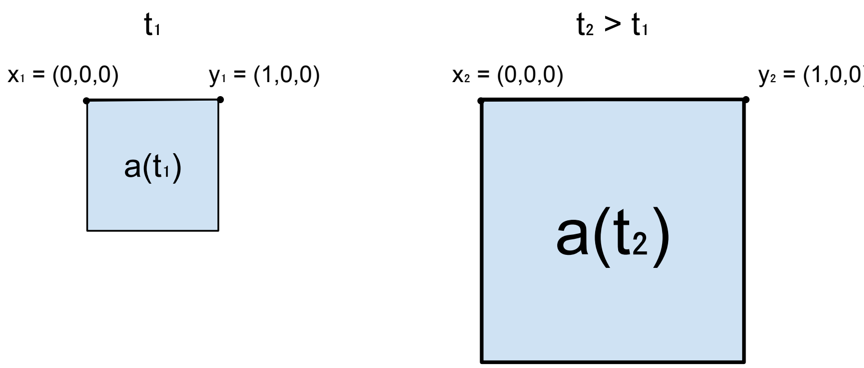

Figure 1.1: Illustration showing the difference between comoving and physical distances. At a time t_1, two points x_1 and y_1 are separated by a distance 1. After some time, at t_2, the Universe has expanded: the scale factor increased and the physical distance between x and y is larger proportionally to the scale factor. In comoving coordinates, however, the distance between x_2 and y_2 remains unchanged.

1.1.2 Cosmological principle

The fundamental concept of the cosmological principle states that the universe is homogeneous and isotropic at large scales. This idea was introduced as a simplification to solve the general relativity equations [5]. It was done by Einstein in 1917 in the case of a static universe [6], but as we saw in previous subsection 1.1.1, it was not relevant any more to assume a static universe after Lemaître and Hubble's works. In 1922, Friedmann showed that a non-static solution for a homogeneous universe is possible [7], quickly followed by evidences of universe expansion in 1927 by Lemaître [1]. Building on these results, Robertson [8] and Walker [9] formulated the general relativistic metric for a homogeneous and isotropic expanding universe. This metric forms the cornerstone of the ΛCDM model. The veracity of this principle is much debated, but latest result from Planck Satellite suggest that the universe is homogeneous and isotropic at 10^{-5} (= 0.001%) at scale above 6° on the sky [10].

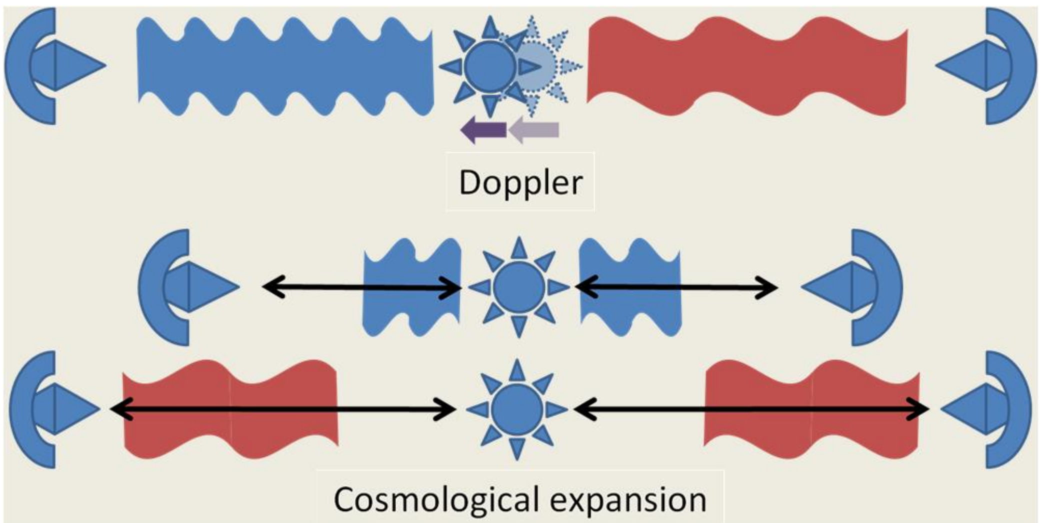

Figure 1.2: Illustration, from [4], showing the difference between Doppler redshift and cosmological redshift. Doppler's redshift [3], on top, is the decrease in wavelength of an object approaching us, and the broadening of an object moving away from us. Cosmological's redshift [2] is the broadening of the wavelength in all directions due to the expansion of the universe.

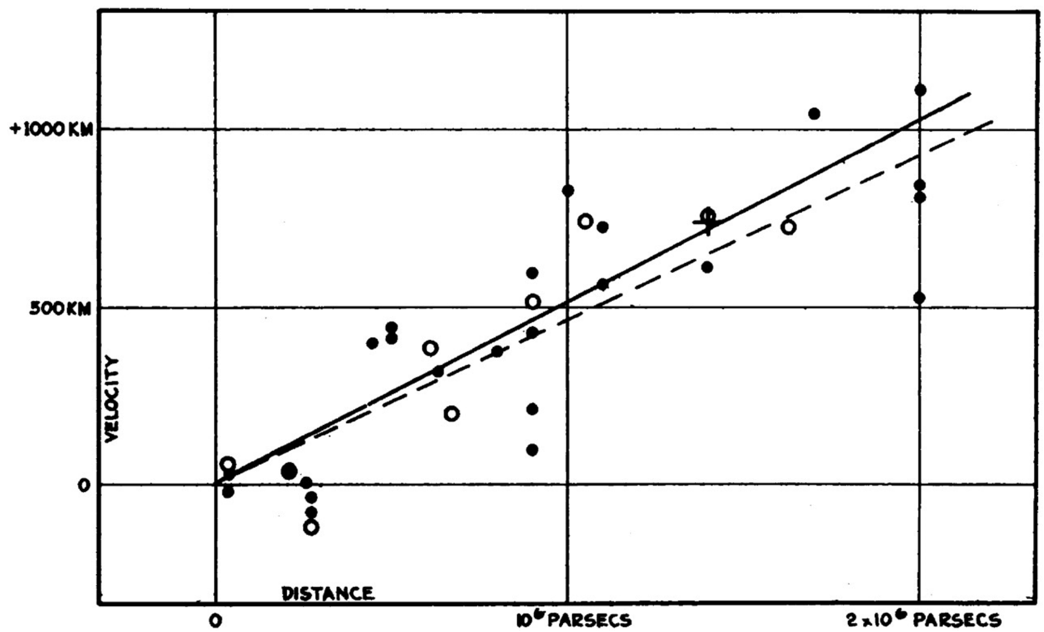

Figure 1.3: Historical velocity-distance diagram from Edwin Hubble [2]. It shows the recession velocity of galaxies as a function of their distance, illustrating the linear relation now known as Hubble's law.

1.1.3 Content of the Universe

In the two previous subsections 1.1.1 and 1.1.2, we introduced the standard assumptions about the universe : it is homogeneous, isotropic and expanding. Now, we now provide a brief overview of its contents. These contents can be classified into three categories: radiation, matter, and the cosmological constant. Each of these components evolves differently with the scale factor and, consequently, with the expansion of the universe. Moreover, each component dominated the universe at different epochs in its history. The evolution of each density is shown in Figure 1.4.

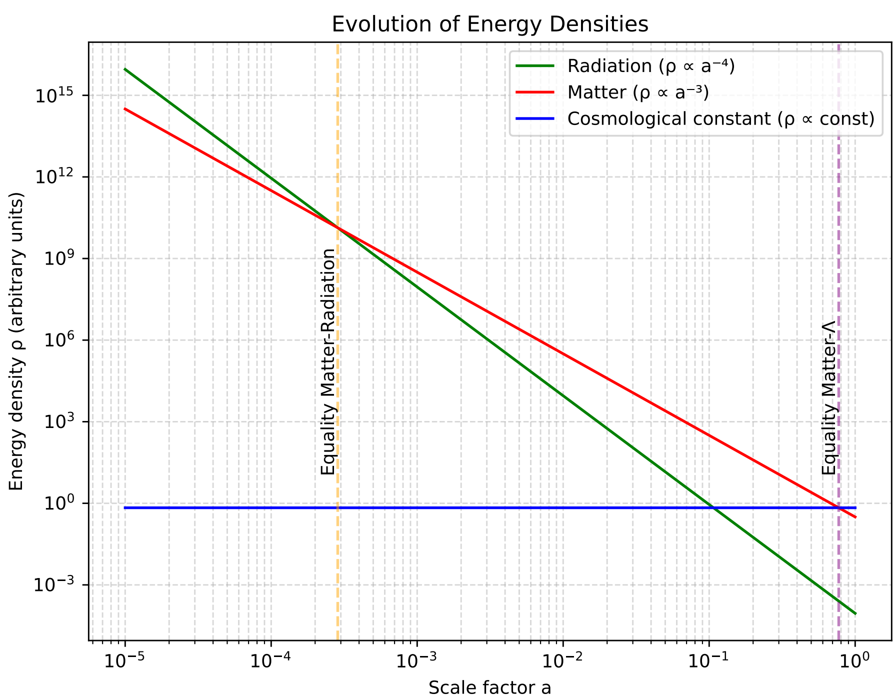

Figure 1.4: Energy density of the Universe's components as a function of the scale factor, highlighting the epochs when matter equals radiation and when matter equals the cosmological constant.

At early times of the universe, the radiation was dominating. This radiation consisted of the relativistic particles : photons and neutrinos, the main contribution being the Cosmic Microwave Background (detailed in Section **??**). The density of radiation and matter were equal at z $\approx$ 3408 according to DESI [11] and z $\approx$ 3387 according to Planck [12], which correspond to approximatively 80000 years after first times of the universe.

Today, radiation contributes about 0.008% of the total energy density, as inferred from Planck 2018 data [12].

As the Universe expanded, non-relativistic matter came to dominate the energy density. This matter is composed of baryonic matter and dark matter. The term "baryonic matter" is misleading, as it designates all non-relativistic particles of the Standard Model; however, its energy density is overwhelmingly dominated by protons and neutrons, hence the terminology. On the other hand, dark matter is a hypothetical form of matter, which only (or mostly) interact through gravitational force, and thus, which does not emit light. It was first suggested by comparing "seen mass" computed from galaxies light and "unseen mass" computed from galaxies velocity in the Coma Galaxies Cluster by Zwicky in 1933 [13]. The mass computed from galaxies velocity was much higher than the luminous mass, suggesting the presence of massive matter that does not emit light.

Since then, multiple independent observations have provided strong evidence for dark matter : galaxy rotation curves [14], the Bullet Cluster through gravitational lensing [15], Baryon Acoustic Oscillations pics observed in CMB [12], formation rate of Large Scale Structures [16], and many others. While we still don't know its nature from the particle point of view, its cosmological properties are well constrained. It is commonly divided into three categories : cold, warm and hot dark matters, which refer to their velocity(or equivalently, whether it was relativistic in the early Universe). This classification determines how far dark matter can propagate (free-stream) and thus influences the formation of structures at different scales. Observations strongly favor the Cold Dark Matter (CDM) scenario [17], in which dark matter is non-relativistic.

Today, Cold Dark Matter (CDM) represents about 26.5% [12] of total energy density, and plays a central role in Standard Model of Cosmology, ΛCDM. The ordinary matter is responsible for 4.5% [12] of the energy density budget, meaning that the energy density of matter occupies 31% of the energy density of the universe, and that 85% of matter in the universe is composes of dark matter.

At z $\approx$ 0.3, corresponding to an age of about 10 Gyr, the Universe transitions from matter domination to dark energy domination, as the cosmological constant becomes the dominant component. The concept of dark energy was first introduced to explain the acceleration of the expansion in the late universe from distant Type Ia supernovae in 1999 [18]. To account for this acceleration, the introduction of a cosmological constant Λ, corresponding to an energy density which does not depend on the scale factor, was needed. Today, dark energy remains one of the greatest unkown in cosmology and in fundamental physics in general.

Dark energy is responsible for 69% of the total energy dansity today, and is by far the main constituent of our universe.

1.1.4 Problems of the Standard Model

From the previous subsection, the Standard Model of cosmology describes the universe as homogeneous, isotropic, expanding, and composed of radiation, matter and dark energy. It is enough to describe the evolution of the universe, and match remarkably well with observations. However, fundamental questions remain unanswered, and in particular three issues cannot be explained within this framework: the horizon problem, the flatness problem, and the monopole problem.

Horizon problem : Why is the CMB temperature so uniform ? As discussed previously, at the time of recombination the Universe was extremely homogeneous, with density fluctuations of order 10^{-5}, and consequently the cosmic microwave background (CMB) exhibits an almost perfectly uniform temperature. This uniformity is naturally explained if the early Universe was in thermal equilibrium, consistent with the fact that the CMB spectrum is an almost perfect blackbody [19]. However, within the standard cosmological model, regions of the CMB that are widely separated on the sky were never in causal contact: light signals would not have had enough time to travel between them since the beginning of the expansion. These regions therefore lie outside each other's particle horizon and cannot have exchanged information or thermalised. This raises a fundamental question: why do causally disconnected regions have the same temperature?

Flatness problem : Why is the universe flat ? In general relativity [5], the geometry of spacetime is determined by its energy content. In a homogeneous and isotropic Universe, this relation is encoded in the Friedmann equations [7], which relate the expansion rate to the total energy density and spatial curvature. From these equations, one defines the critical density, ρ_c, which corresponds to a spatially flat Universe. Depending on the ratio $\Omega = \rho/\rho_c$, the Universe can be:

$$\begin{aligned}
\Omega = 1 &\quad : \text{flat (Euclidean geometry)},\\
\Omega > 1 &\quad : \text{closed (positive curvature)},\\
\Omega < 1 &\quad : \text{open (negative curvature)}.
\end{aligned}$$

Current measurements, notably from the Planck mission [12], indicate that Ω differs from unity by less than about one percent, implying that the Universe is extremely close to spatial flatness. However, within the standard Friedmann evolution, flatness is not a stable state: any small deviation from $\Omega = 1$ grows with time. Consequently, evolving backward in time implies an extreme fine-tuning of the initial condition. For example, a deviation of order $|\Omega_0 - 1| \sim 10^{-3}$ today corresponds to an initial deviation of order 10^{-60} near the Planck epoch. This extraordinary sensitivity suggests an apparent fine-tuning problem: why was the early Universe so precisely close to flatness?

Monopole problem : Why are magnetic monopoles not observed ? Grand unified theories (GUTs), which attempt to unify the electromagnetic, weak, and strong interactions at high energy scales, generically predict the existence of topological defects formed during phase transitions in the early Universe. Among these relics are magnetic monopoles, hypothetical particles carrying an isolated magnetic charge, whose existence was first proposed by Dirac in 1931 [20]. In standard cosmology, such monopoles should be efficiently produced during the spontaneous symmetry breaking associated with the GUT phase transition. Causally disconnected regions of space choose different vacuum states, leading to the formation of stable topological defects. As a consequence, one expects a large relic abundance of monopoles. However, monopoles are not observed in the present Universe, despite extensive experimental searches. This discrepancy constitutes the monopole problem: standard cosmology predicts a relic density of monopoles that is many orders of magnitude larger than observational bounds. Should I talk about higher order topological defects ? Like cosmic cords, walls, ...

Finally, an additional issue remains throughout the Standard Model of Cosmology's equations. As with any set of equations describing the evolution of a physical system, the dynamics are only fully specified once appropriate initial conditions are given. This naturally raises the question: how can the initial conditions of our Universe be determined or constrained observationally? An answer to the three problems and to this issue was proposed by Alan Guth in 1981, through the idea of an exponential expansion phase at the very beginning of the Universe [21]. The model was then developed by Andrei Linde in 1983 [22], and has since given rise to a wide variety of models. We will explain in the following subsection the idea behind this theory and how it can solve the issues discussed previously.

1.1.5 Inflation theory

Should I talk about the Inflation scale ($\exp -N$, with $N \sim 60$), to make more understandable that it solves all the problems

The theory of Inflation postulates that in the very early universe, all regions were causally connected. These regions were then disconnected due to an exponentially expanded phase. This theory was developed to solve the various problems presented previously 1.1.4. **Horizon Problem.** Indeed, if all regions were connected in the very early universe, they could have thermalised, explaining why the CMB temperature is uniform at a very large scale. **Flatness Problem.** As we said, a nearly flat universe is an unstable solution, meaning that it is very unlikely to happen. But, Inflation suggests that they do not care about the curvature before it occurs, as an exponential expansion will make all local curvatures negligible. We can understand this argument with the analogy of a sphere: if we zoom enough on a sphere, its surface will be flat. This mechanism explains why the universe is flat today, independently of its initial conditions. **Monopole Problem.** The GUT theories predict the formation of monopoles, but the exponential expansion dilutes their density to the point where they become unobservable today.

Additionally, we talked about all the properties of the universe for the Λ-CDM model: homogeneity, isotropy, and flat geometry. These initial conditions seem too "fine-tuned". Inflation also solves this issue by making any specific initial condition negligible due to the exponential expansion. To go even further, Inflation would have expanded quantum fluctuations to a macroscopic scale, responsible for the inhomogeneity of density observed in the temperature fluctuations in the CMB, which led to the large-scale structures' formation under gravitational effects today.

The most widely studied inflationary scenario involves a scalar field that slowly evolves toward its true ground state [23] [24]. In this framework, the field's potential energy remains nearly constant and therefore rapidly comes to dominate over both its kinetic energy and the energy density of other components. Inflation comes to an end when the field reaches the minimum of its potential, after which it undergoes oscillations and decays into lighter particles, eventually producing those of the Standard Model of Particles. A wide variety of potential shapes have been proposed in the literature (see [25] for an overview of the different models), and many of these can be adjusted to fit current observations. This highlights the importance of observational constraints in discriminating between competing inflationary models.

Is it enough on Inflation ? Or should I talk more about the mechanism, slow-roll approximation, ...?

1.2 Thermal history of the universe

In this section, we aim to produce a quick overview of the history of the universe, discussing the key milestones of its evolution and their implication.

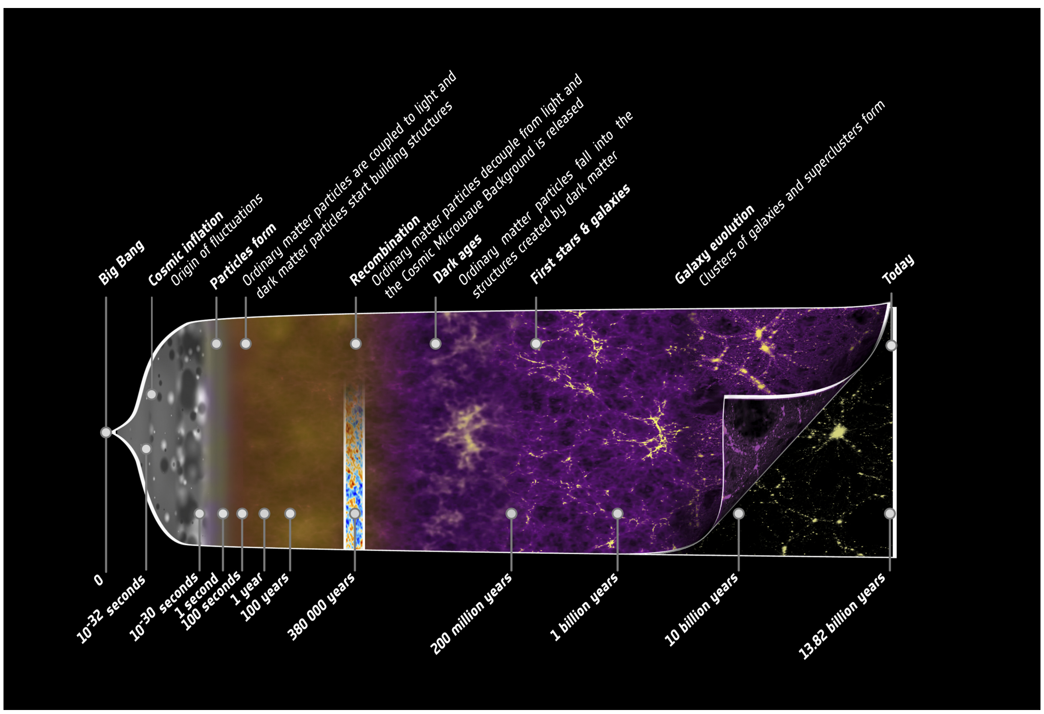

Figure 1.5: Illustration of the history of the universe from the early beginning to today, highlighting the key events discuss in this section. Credit: ESA – C. Carreau.

1.2.1 Inflation and First particles

Is it useful ? Should I add temporal and/or energetic scale(s) ?

As discussed in Section 1.1.5, the standard cosmological model predicts an early phase of accelerated, quasi-exponential expansion known as Inflation. This paradigm addresses several major issues in cosmology and provides a mechanism for setting the initial conditions of the observable Universe. At the end of inflation, the scalar field driving this expansion reached its minimum potential and decays into lighter particles as the universe cool down. This process, known as reheating, initiates the production of the particle content of the Standard Model through a cascade of interactions and decays. Subsequently, matter and antimatter are generated during baryogenesis, with a small asymmetry in favor of matter. Their annihilation produces numerous photons, leaving behind a residual matter component due to this asymmetry. As the temperature continues to decrease, the fundamental interactions—strong, weak, and electromagnetic—become distinct through a sequence of symmetry-breaking transitions. This leads to the formation of hadrons, primarily protons and neutrons. Neutrinos eventually decouple from the primordial plasma, forming a relic background known as the *cosmic neutrino background*. Finally, universe's temperature becomes too low for electron/position pairs production. The remainings electron and positron annihilate to produce photons. The result is a homogeneous universe dominated by radiation (photons, neutrinos) [26].

Is citation OK ? I just want to cite a general reference for this but don't know how to do

1.2.2 Big Bang Nucleosynthesis

At early times, the temperature of the Universe was too high for nuclei to form, as any bound state was immediately broken by high-energy photons. As the Universe expanded and cooled, this process became inefficient, allowing nuclear reactions to proceed.

Weak interactions froze out at a temperature of $T \sim 0.8\,\mathrm{MeV}$ ($t \sim 1\,\mathrm{s}$), fixing the neutron-to-proton ratio at $n/p \sim 1/6$, which subsequently decreased due to free neutron decay. Big Bang Nucleosynthesis (BBN), first introduced in [27], effectively began when the temperature dropped below

$T \sim 0.1\,\mathrm{MeV}$ ($t \sim 200\,\mathrm{s}$), enabling the formation of deuterium through

$$p + n \rightarrow {}^{2}\mathrm{H} + \gamma\,. \tag{1.2}$$

This marked the end of the so-called *deuterium bottleneck*. Rapid nuclear reactions then led to the synthesis of light elements, primarily ^{3}He, ^{4}He, and trace amounts of ^{7}Li. Due to its large binding energy, ^{4}He is the most stable and abundant product after hydrogen, with a final mass fraction of about 25%, while the remaining baryonic matter consists mostly of hydrogen.

The predicted primordial abundances are in good agreement with observations, making BBN one of the key pillars of the standard cosmological model [28]. The Figure 1.6 shows the evolution of abundances of the lighter elements with time and energy.

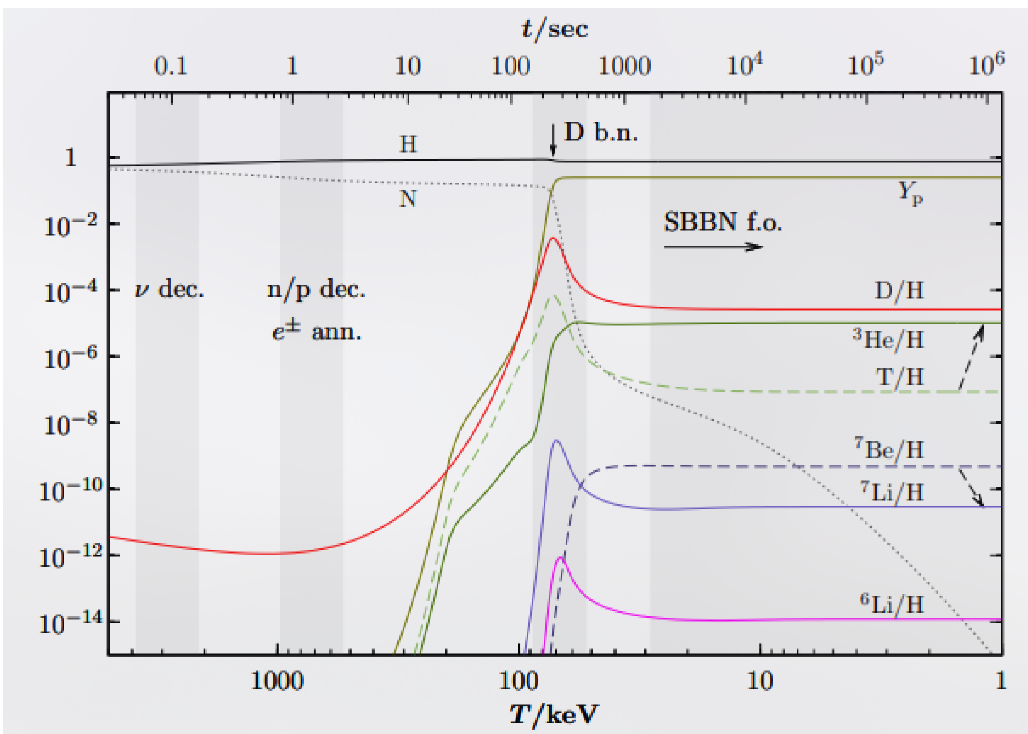

Figure 1.6: Illustration of the Big Bang Nucleosynthesis process, showing the abundances of elements with time (top-axis) and energy (bottom-axis). Taken from [29].

1.2.3 Baryon Acoustic Oscillations

During the radiation-dominated era, photons were tightly coupled to baryons through Thomson scattering of free electrons [26], forming a single photon-baryon fluid. Due to the primordial density fluctuations generated during inflation (see Section 1.1.5), overdense regions acted as gravitational potential wells, attracting baryonic matter. However, photon pressure counteracted gravitational fall. The competition between gravity and radiation pressure led to acoustic oscillations in the coupled photon-baryon fluid, generating spherical sound waves propagating outward from initial overdensities. These oscillations ceased at recombination (see Section 1.2.4), when photons decoupled from baryons and began to free-stream. The maximum distance traveled by these sound waves defines a characteristic scale, known as the sound horizon. This scale is imprinted in the late-time matter distribution as a preferred clustering scale. Baryon Acoustic Oscillations (BAO) were first detected in the large-scale distribution of galaxies by the Sloan Digital Sky Survey [30] and independently confirmed by the 2dF Galaxy Redshift Survey [31]. They appear as a distinct peak in the two-point correlation function at a scale of approximately 150 Mpc, corresponding to the sound horizon at recombination (see Fig. 1.7).

Additionally, dark matter behaved differently, as it does not interact with photons. Consequently, dark matter is affected only by gravity, creating overdensities that will be the foundation of Large Scale Structures 1.2.5.

1.2.4 Recombination

Recombination refers to the epoch when the Universe became sufficiently cool and dilute that photons could no longer efficiently maintain the coupling between matter and radiation, leading to the decoupling of the primordial plasma and the transparency of the universe. From this moment on, photons propagate freely through space, forming a relic radiation known as the *Cosmic Microwave Background* (CMB), discussed in detail in Section 1.3. During this transition, nuclei were no longer kept ionised by high-energy photons, allowing the formation of the first neutral atoms, mainly hydrogen and helium (see Section 1.2.2).

Prior to recombination, matter and radiation were tightly coupled, and the primordial plasma was in thermal equilibrium. As a consequence, the CMB is expected to exhibit an almost perfect blackbody spectrum, described

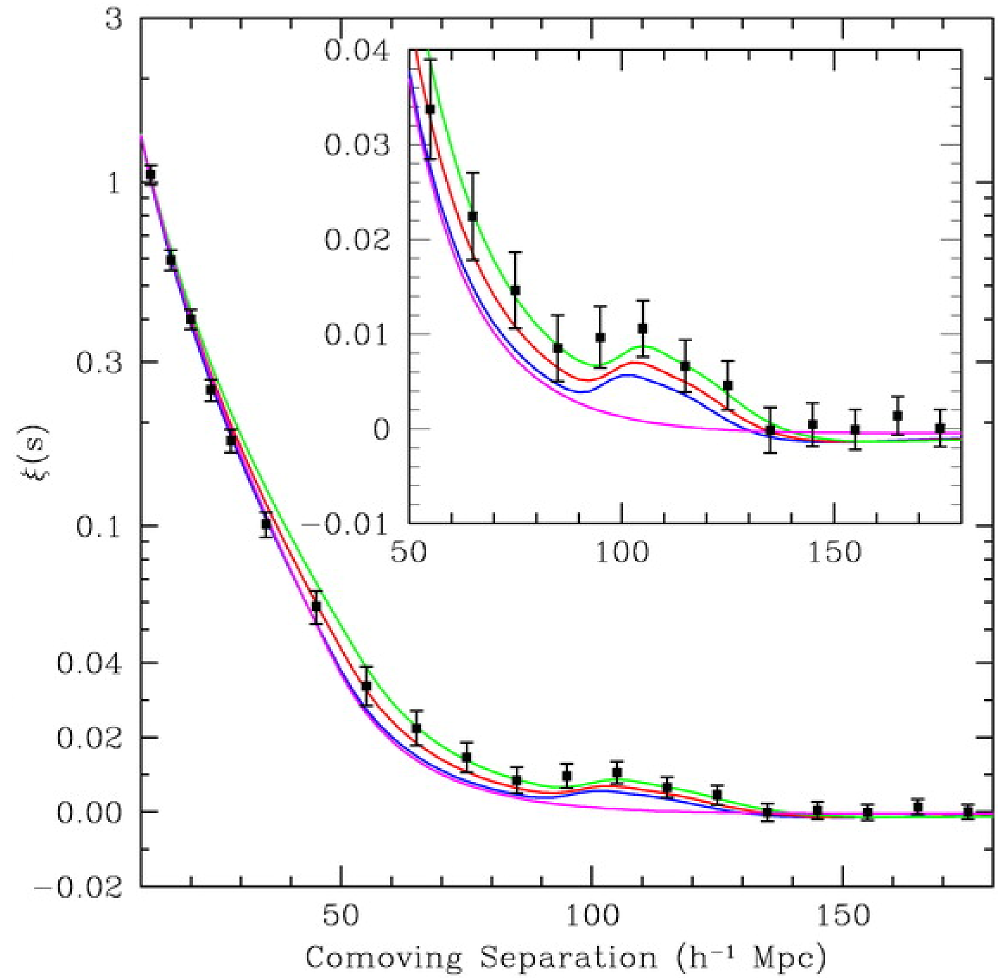

Figure 1.7: Matter two-point correlation function over comoving distance of the SDSS LRG sample. The BAO peak is visible at 150 Mpc, corresponding to an overdensity of matter at this distance. From [30].

by Planck's law [32]:

$$B_\nu(T) = \frac{2h\nu^3}{c^2}\frac{1}{e^{h\nu/kT}-1}. \tag{1.3}$$

This prediction has been confirmed by the COBE satellite, which provided the most precise measurement of a blackbody spectrum to date, as shown in Figure 1.8.

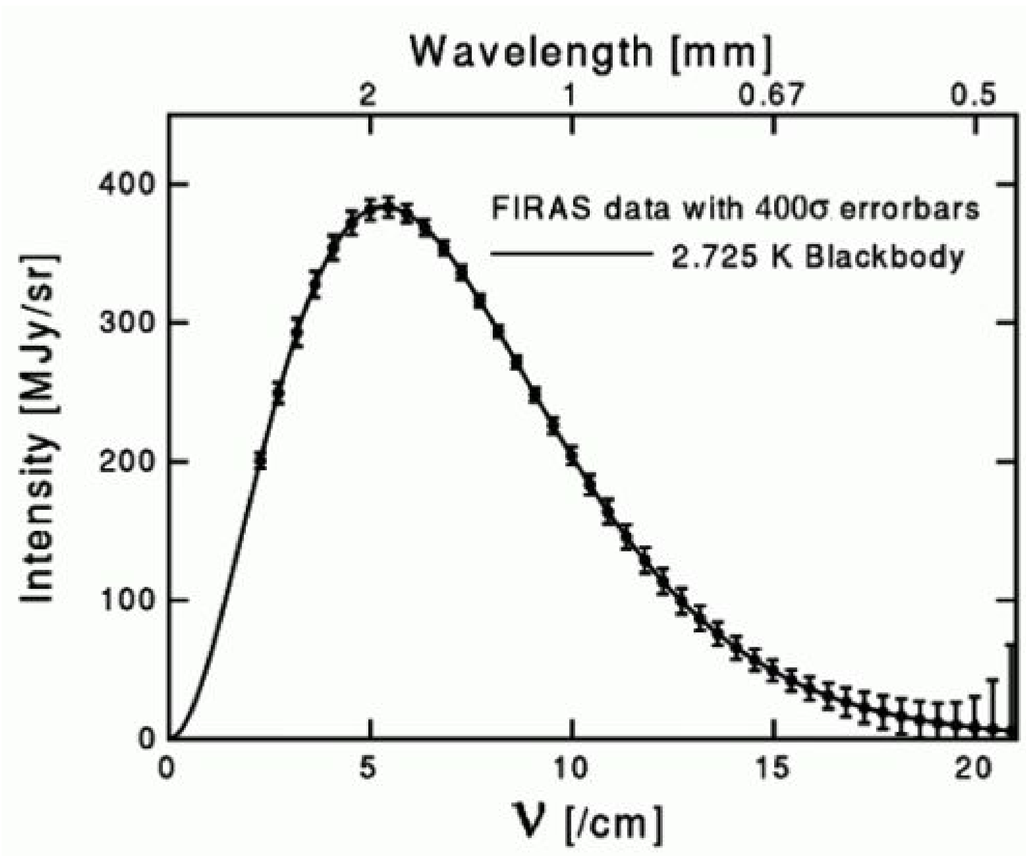

Figure 1.8: CMB spectrum measured by FIRAS instrument on COBE satellite. Plain line is the theoretical spectrum for a blackbody at 2.725 K, points are data from FIRAS with errorbars multiplied by 400 to be visible.

1.2.5 Large Scale Structures

After matter–radiation decoupling, baryonic matter evolves primarily under the influence of gravity. The small primordial density perturbations grow through gravitational instability. In the linear regime, their growth is proportional to the scale factor during matter domination. Dark matter, having decoupled earlier, forms gravitational potential wells into which baryons subsequently fall, leading to the formation of structures. This hierarchical process results in the emergence of the large-scale structure of the Universe, including halos, galaxies, clusters, and filaments. On large scales, the statistical properties of matter fluctuations are described by the matter power spectrum $P(k)$, which encodes both the primordial fluctuations and their subsequent evolution through cosmic expansion and gravitational collapse. A key observable imprint of these processes is given by Baryon Acoustic Oscillations (BAO), which provide a standard ruler in the distribution of matter (see Section 1.2.3).

1.3 Cosmic Microwave Background

So far, we have repeatedly referred to the Cosmic Microwave Background (see 1.2.4). However, as the CMB constitutes our primary observational probe of the primordial Universe and lies at the core of this thesis, we now present a more detailed discussion of its physical origin, its specific properties, and its connection to cosmological parameters, both through its intensity and polarization signals.

1.3.1 CMB Temperature

As discussed above (see 1.2.4), the CMB follows a nearly perfect blackbody spectrum. Its emission is therefore entirely determined by a single parameter: its temperature. Since the temperature of the photon–baryon plasma at recombination depends on local physical conditions, the observed CMB temperature anisotropies directly trace spatial fluctuations in this plasma. For this reason, CMB intensity anisotropies are conventionally expressed in terms of an effective temperature, which encodes information about the density fluctuations of the primordial plasma and, more generally, about its underlying physics.

Temperature Monopole and Dipole The temperature monopole of the CMB is its mean temperature across the sky. It has been first measured by FIRAS instrument of COBE satellite at $T_0 = 2.7255 \pm 0.0006K$ [19]. This measurement remains the most precise determination of the CMB monopole, as subsequent experiments have primarily focused on measuring anisotropies around this mean temperature. From particle physcis considerations, photon–baryon decoupling occurs when the primordial plasma reaches a temperature $T_{rec} \approx 3000K$. As discussed in 1.1.1, cosmic expansion induces a redshift of photon wavelengths. For a blackbody spectrum, this implies that the temperature scales with redshift as :

$$T(z) = T_0(1 + z), \tag{1.4}$$

where T_0 is the present CMB temperature and z its redshift. Using the measured value of T_0 and the computed recombination temperature T_{rec}, one can infer that the CMB was emitted at a redshift $z_{\text{rec}} \approx 1100$.

A first level of anisotropy is induced by the Earth's motion with respect to the CMB rest frame. Indeed, because the Earth has a peculiar velocity relative to the CMB rest frame, arising from the motion of the Solar System and of the Milky Way, the CMB photon wavelengths (and thus the observed temperature) are Doppler shifted due to the electromagnetic Doppler effect [3]. The CMB solar dipole was first observed by the DMR instrument aboard the COBE satellite [33], and later measured more accurately by WMAP [34] and Planck [35]. The solar dipole, simulated using Python Sky Model [1] (PySM) is shown in Figure 1.9.

CMB Anisotropies As the primordial plasma density has small density inhomogeneities, at the level of $\approx 1e^{-5}$, we except to detect them in the temperature map of the CMB. The first observation of these anisotropies was, once again, done by COBE satellite and its DMR instrument [33]. The frequency sky maps of DMR after 4 years of observation are shown figure 1.10.

Angular Power Spectrum

Cosmological Parameters

[1] `https://github.com/galsci/pysm`

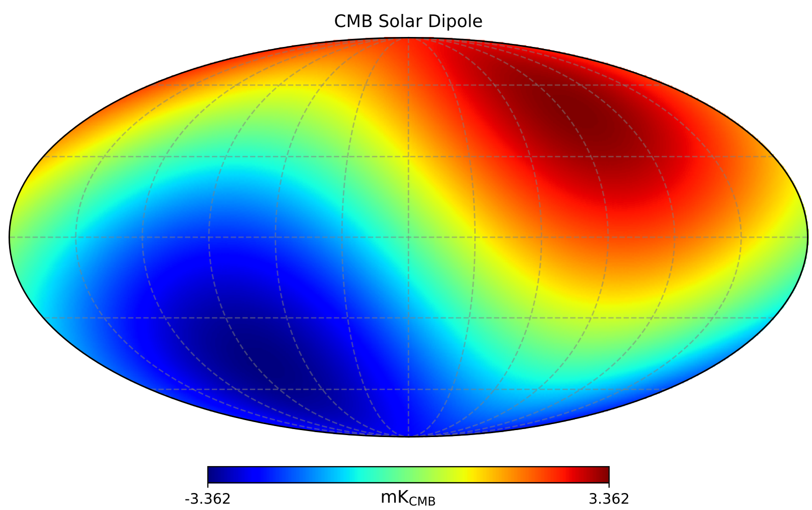

Figure 1.9: CMB solar dipole simulated using the PySM software [36] and calibrated using Planck 2018 measurements [35].

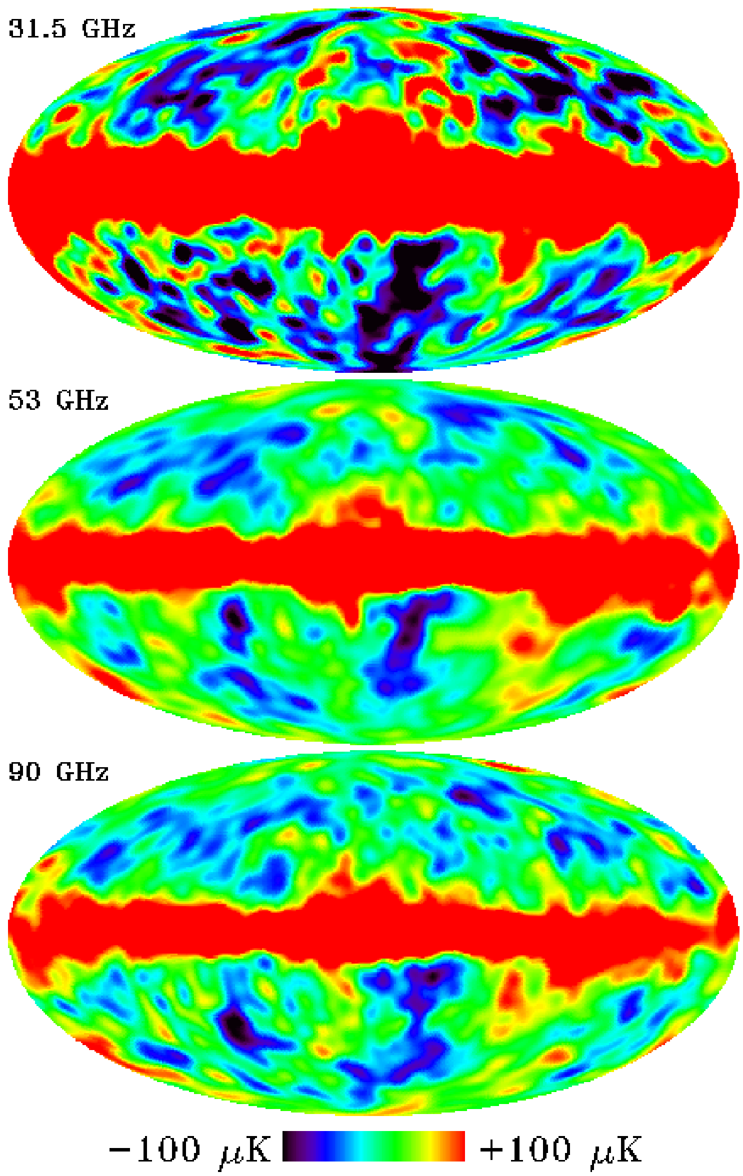

Figure 1.10: Full-sky maps of DMR instrument of COBE satellite after 4 years of observation, from the 31.5, 53 and 90 GHz channels. Monopole and Dipole emissions have been removed. The bright red bands at the equator of all maps are due to the galactic emissions, while temperature anisotropies are visible above and below. Maps taken from [37].

1.3.2 CMB Polarisation

Reminder on polarisation

Compton Scattering

Gravitational Waves

Power Spectrum

Cosmological parameters

1.4 CMB contaminations

1.4.1 Lensing

1.4.2 Astrophysical Dust

1.4.3 Synchrotron

1.4.4 Atmosphere

1.5 Past & Future Experiments

1.5.1 Past Experiements

1.5.2 Current Experiments

1.5.3 Future Experiments

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